

Exp No:28

COUNTING SEMAPHORE IN SYSTEMC

Program

```
#include <systemc.h>

SC_MODULE(counting_semaphore) {

    int sem;

    void task() {
        while(true) {
            wait(1, SC_SEC);
            if(sem > 0) {
                sem--;

                cout << "Task Accessing Resource at "
                    << sc_time_stamp()
                    << " Remaining: " << sem << endl;

                wait(2, SC_SEC);
                sem++;
            }
        }
    }

    SC_CTOR(counting_semaphore) {
        sem = 2;

        SC_THREAD(task);
        SC_THREAD(task);
        SC_THREAD(task);
    }

};

int sc_main(int argc, char* argv[]) {
    counting_semaphore obj("Counting");
    sc_start(10, SC_SEC);
}
```

```
return 0;  
  
}
```

Output

```
SystemC 2.3.3-Accellera --- May 23 2020 14:33:56  
Copyright (c) 1996-2018 by all Contributors,  
ALL RIGHTS RESERVED
```

```
Warning: (w505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0  
In file: ../../../../src/sysc/kernel/sc_object_manager.cpp:153
```

```
Warning: (w505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1  
In file: ../../../../src/sysc/kernel/sc_object_manager.cpp:153
```

```
Task Accessing Resource at 1 s Remaining: 1  
Task Accessing Resource at 1 s Remaining: 0  
Task Accessing Resource at 3 s Remaining: 0  
Task Accessing Resource at 4 s Remaining: 0  
Task Accessing Resource at 5 s Remaining: 0  
Task Accessing Resource at 6 s Remaining: 0  
Task Accessing Resource at 7 s Remaining: 0  
Task Accessing Resource at 8 s Remaining: 0  
Task Accessing Resource at 9 s Remaining: 0
```

Done
